Worksheet 5-10 & 5-11 B Optimization Problems #2
AP Calculus Name_____

Solve each problem.

1.) Find the point on the line y = 2x + 3 that is closest to the origin.

2.) Find the point on the curve y = $\sqrt{x}$ that is closest to the point (3, 0).

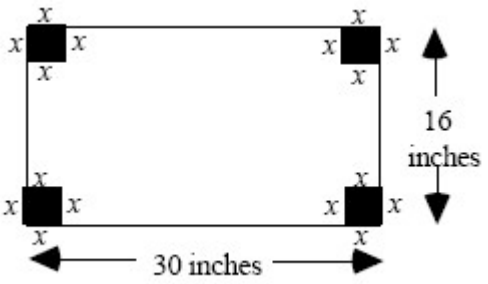
3.) Find two numbers whose difference is 100 and whose product is a minimum.

4.) The sum of two positive numbers is 16. What is the smallest possible value of the sum of their squares?

5.) Find the dimensions of a rectangle with an area of 1000 m^2 whose perimeter is as small as possible.

6.) A farmer wants to fence in an area of 1.5 million square feet in a rectangular field and then divide it in half with a fence parallel to one of the sides of the rectangle. How can he do this so as to minimize the cost of the fence?

7.) An open box is to be made from a piece of metal 16 by 30 inches by cutting out squares of equal size from the corners and bending up the sides. What size square should be cut out to create a box with greatest volume? What is the maximum volume?



8.) A box with a square base and open top must have a volume of 32,000 cm^3. Find the dimensions of the box that minimize the amount of material used.

9.) The top and bottom margins of a poster are each 6 cm and the side margins are each 4 cm. If the area of printed material on the poster is fixed at 384 cm^2, find the dimensions of the poster with the smallest area.

Answers:

1.) $(-1.2, 0.6)$ 2.) $\left(\frac{5}{2}, \sqrt{\frac{5}{2}}\right)$ 3.) –50 and 50 4.) 128

5.) $\frac{1000}{\sqrt{1000}}$ m x 1000 m or 31.623 m x 31.623 m

6.) 1000 ft by 1500 ft with the fence parallel to the short side.

7.) the square should be $\frac{10}{3}$ in x $\frac{10}{3}$ in; the volume is $\frac{19600}{27}$ in^3 or 725.926 in^3

8.) 40 cm x 40 cm x 20 cm

9.) 24 cm x 36 cm